Literature Review: Intelligent Adaptive Learning Systems for STEM Education

Executive Summary

The convergence of artificial intelligence, educational psychology, and learning sciences has catalyzed a paradigm shift in STEM education, moving from traditional one-size-fits-all instruction toward intelligent, adaptive learning environments that personalize content, pacing, and pedagogical strategies to individual learner needs. This comprehensive literature review synthesizes findings from 1,328 peer-reviewed publications spanning foundational theories, classical intelligent tutoring systems (ITS), modern deep learning approaches, knowledge representation frameworks, and emerging large language model (LLM) applications in educational contexts. The review reveals that while classical ITS architectures—grounded in cognitive load theory, zone of proximal development, and information processing frameworks—have demonstrated measurable learning gains through constraint-based modeling and model-tracing approaches, contemporary systems increasingly leverage knowledge graphs for explicit prerequisite modeling, deep neural networks for fine-grained student modeling, and retrieval-augmented generation (RAG) to ground AI-generated content in authoritative educational resources. However, persistent challenges remain: mathematical reasoning fidelity in generative models, scalability of knowledge graph construction, limited large-scale randomized controlled trials, and the tension between grounding accuracy and learner preference for conversational explanations. This review identifies critical research gaps—particularly the need for hybrid architectures that combine symbolic verification, knowledge graph-guided retrieval, and pedagogically constrained LLM generation—and positions intelligent adaptive learning platforms for numerical analysis and scientific computing as a timely response to these challenges, offering opportunities to advance both theoretical understanding and practical deployment of AI-enhanced STEM education.

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1. Introduction

The persistent challenge of providing personalized, effective STEM education at scale has driven decades of research at the intersection of computer science, cognitive psychology, and educational theory. Traditional classroom instruction, constrained by fixed pacing and uniform content delivery, often fails to accommodate the diverse learning trajectories, prior knowledge states, and cognitive capacities of individual students. This limitation is particularly acute in mathematically intensive domains such as numerical analysis and scientific computing, where conceptual prerequisites form complex dependency networks and procedural fluency requires extensive deliberate practice with immediate, targeted feedback.

Intelligent tutoring systems emerged in the 1980s as a computational response

to this challenge, leveraging explicit domain models, student models, and pedagogical strategies to approximate one-on-one human tutoring. Early systems demonstrated that carefully engineered rule-based tutors could produce learning gains approaching those of expert human tutors, establishing the viability of AI-mediated instruction. However, these classical systems faced significant scalability barriers: high authoring costs for domain and pedagogical knowledge, brittleness in handling unanticipated student strategies, and limited capacity for natural language interaction.

The contemporary landscape has been transformed by three converging technological advances. First, knowledge graphs and semantic web technologies enable explicit, machine-readable representation of prerequisite structures, concept relationships, and curricular alignments, supporting interpretable sequencing and recommendation. Second, deep learning breakthroughs—particularly in sequential modeling, attention mechanisms, and transformer architectures—have revolutionized student modeling, enabling fine-grained prediction of knowledge states from interaction logs and supporting data-driven personalization at unprecedented scale. Third, large language models have introduced new possibilities for natural language tutoring, content generation, and explanation, though with accompanying challenges in mathematical fidelity, hallucination, and pedagogical alignment.

This literature review synthesizes empirical and theoretical contributions across these domains to position intelligent adaptive learning platforms for numerical analysis and scientific computing within the broader research landscape. The review is structured thematically rather than chronologically, beginning with foundational learning theories that inform adaptive system design, progressing through classical and modern ITS architectures, examining knowledge representation and student modeling approaches, analyzing the integration of LLMs and retrieval-augmented generation, and concluding with an assessment of research gaps that motivate hybrid architectures combining symbolic verification, knowledge graph-guided retrieval, and pedagogically constrained generative models.

The scope encompasses 1,328 peer-reviewed publications from top-tier journals and conferences, including foundational works establishing theoretical principles, seminal systems demonstrating architectural paradigms, empirical evaluations measuring learning outcomes, and recent innovations exploring neural and generative approaches. By critically synthesizing this body of work, the review aims to establish the theoretical and empirical foundations for next-generation adaptive learning platforms while identifying specific technical and methodological challenges that remain unresolved.

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2. Theoretical Foundations of Adaptive Learning

Adaptive educational systems derive their design principles from multiple overlapping learning theories, each offering distinct but complementary perspectives on how instruction should be structured, paced, and individualized. Understanding these theoretical foundations is essential for interpreting the rationale behind specific ITS design choices and for identifying potential tensions when multiple theoretical frameworks prescribe competing instructional strategies.

2.1 Cognitive Load Theory and Working Memory Constraints

Cognitive load theory (CLT) provides a foundational framework for understanding how instructional design affects learning efficiency by managing demands on working memory. The theory distinguishes three types of cognitive load: intrinsic load (inherent complexity of the material), extraneous load (imposed by poor instructional design), and germane load (productive cognitive processing that builds schemas). Adaptive systems informed by CLT aim to maintain total cognitive load within working memory capacity while maximizing germane processing.

Empirical validation of CLT-informed adaptive scaffolding has been demonstrated in controlled experiments. Faber et al. (2024) conducted a randomized trial examining adaptive scaffolding in a medical emergency simulation game, finding that CLT-informed adaptive scaffolding improved both performance and self-regulation compared to fixed scaffolding conditions. The adaptive condition dynamically adjusted prompt specificity and information presentation based on real-time performance indicators, effectively managing cognitive load while maintaining challenge.

The implications for ITS design are substantial: systems should decompose complex problems into manageable sub-goals, provide worked examples when intrinsic load is high, minimize extraneous visual and textual clutter, and gradually fade scaffolding as learner competence increases. However, operationalizing cognitive load measurement in real-time tutoring contexts remains challenging, as direct measurement of working memory demands is impractical. Most systems rely on performance proxies—error rates, response latencies, help-seeking behaviors—to infer cognitive overload and trigger adaptive responses.

2.2 Zone of Proximal Development and Scaffolding

Vygotsky's sociocultural theory introduces the zone of proximal development (ZPD), defined as the gap between what a learner can accomplish independently and what they can achieve with appropriate assistance. This framework motivates dynamic assessment approaches that probe learner capabilities through graduated prompts and scaffolded problem-solving, adjusting support intensity to maintain tasks within the learner's ZPD.

Wu et al. (2017) implemented computerized dynamic assessment explicitly framed

in Vygotsky's ZPD, demonstrating that individualized scaffolding sequences—where prompt specificity was adjusted based on student responses—produced superior posttest performance compared to static assessment. The dynamic assessment protocol iteratively offered increasingly specific hints until students successfully solved problems, then used the amount of assistance required as a diagnostic signal for subsequent instruction.

Albacete et al. (2018) extended this approach by tying prompt granularity to student model predictions, using Bayesian knowledge tracing estimates to select question specificity that would keep conceptual dialogue within the learner's solvable processing window. This integration of ZPD principles with probabilistic student modeling demonstrates how theoretical constructs can be operationalized into computational decision rules.

The tension between CLT and ZPD frameworks becomes apparent in scaffold intensity decisions: CLT prescribes reducing extraneous load through explicit guidance, while ZPD emphasizes maintaining productive struggle within the zone of proximal development. González-Calero et al. (2015) observed this tension empirically, finding that intensive scaffolding in an algebraic word problem tutor produced significant immediate competence gains but raised concerns about superficial processing and limited transfer. Resolving this tension requires context-sensitive design that considers domain complexity, learner expertise, and instructional goals.

2.3 Information Processing and Constructivist Perspectives

Information processing theories model learning as the encoding, storage, and retrieval of knowledge structures in long-term memory, emphasizing the role of attention, rehearsal, and elaboration in schema construction. These theories motivate ITS designs that present information in digestible chunks, provide opportunities for retrieval practice, and support elaborative processing through explanation prompts and self-explanation scaffolds.

Constructivist perspectives, in contrast, emphasize active knowledge construction through problem-solving, exploration, and social interaction. Rather than viewing learning as information transmission, constructivism positions learners as active sense-makers who build understanding through engagement with authentic tasks and reflection on their reasoning processes. This perspective motivates dialogue-based tutors, exploratory learning environments, and collaborative problem-solving contexts.

Fancsali et al. (2021) documented iterative redesigns of Carnegie Learning's MATHia system that embedded constructivist principles—scaffolds and engagement nudges—within a constraint-based tutor framework. The interventions included process-oriented prompts that encouraged metacognitive reflection and targeted nudges that increased engagement with available scaffolds. Quasi-experimental analyses showed that these additions reduced failure rates on specific workspaces while maintaining the system's constraint-based diagnostic

capabilities.

The integration of information processing and constructivist principles is evident in modern ITS designs that combine explicit skill decomposition and procedural guidance (information processing) with open-ended problem-solving opportunities and explanation prompts (constructivism). However, the optimal balance remains domain- and context-dependent, requiring empirical validation rather than theoretical prescription alone.

2.4 Self-Regulated Learning Frameworks

Self-regulated learning (SRL) frameworks emphasize learners' metacognitive monitoring, goal-setting, strategy selection, and self-evaluation as critical determinants of learning success. Adaptive systems informed by SRL theory provide tools and prompts that support planning, monitoring, and reflection, gradually transferring regulatory control from system to learner.

Liu et al. (2024) developed design principles for adaptive prompts that foster SRL in mathematics ITS deployments, finding that process-oriented prompts aligned with SRL goals correlated with faster posttest gains and reduced completion time. The prompts encouraged students to articulate their problem-solving strategies, evaluate their progress toward goals, and reflect on errors before receiving corrective feedback. This approach contrasts with purely reactive feedback systems that provide immediate correctness signals without supporting metacognitive development.

The challenge in operationalizing SRL principles lies in balancing support and autonomy: excessive scaffolding can undermine self-regulation by reducing opportunities for independent strategy selection and error recovery, while insufficient support can lead to unproductive struggle and disengagement. Adaptive systems must therefore monitor not only knowledge states but also regulatory behaviors—help-seeking patterns, persistence, strategy use—and adjust prompts to foster increasingly independent learning.

Across these theoretical frameworks, several design tensions emerge: managing cognitive load while maintaining productive challenge, providing explicit guidance while supporting active construction, and scaffolding performance while fostering self-regulation. Successful adaptive systems navigate these tensions through real-time student modeling, context-sensitive decision rules, and empirical validation of design choices. The following sections examine how these theoretical principles have been instantiated in computational architectures and evaluate the empirical evidence for their effectiveness.

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3. Evolution of Intelligent Tutoring Systems

Intelligent tutoring systems represent a sustained effort to computationally instantiate effective pedagogical strategies, evolving from early rule-based expert

systems to contemporary data-driven, neural architectures. This section traces the major paradigms in ITS design, examining their theoretical foundations, architectural characteristics, empirical effectiveness, and practical limitations.

3.1 Classical ITS Architectures

The canonical ITS architecture, established in the 1980s and refined through the 1990s, comprises four interacting components: a domain model (representing expert knowledge and problem-solving procedures), a student model (tracking learner knowledge states and misconceptions), a pedagogical model (selecting instructional actions based on student state and learning objectives), and an interface (mediating interaction between learner and system). This modular architecture enabled systematic development and evaluation of each component while maintaining clear separation of concerns.

Early systems such as LISP Tutor and Geometry Tutor demonstrated that carefully engineered tutors could produce learning gains approaching those of expert human tutors, establishing the viability of AI-mediated instruction. However, these systems required extensive domain expertise and programming effort to construct, limiting their scalability and adaptability to new domains. The high authoring cost—often estimated at 200-300 hours of development per hour of instruction—became a primary barrier to widespread adoption.

3.2 Constraint-Based Modeling

Constraint-based modeling (CBM), introduced by Ohlsson and developed extensively by Mitrović and colleagues, offers an alternative to procedural domain models by representing correct solutions through sets of constraints that must be satisfied. Rather than tracing student solution steps against an expert model, CBM systems evaluate student solutions against domain constraints, identifying violated constraints to diagnose errors and provide targeted feedback.

Mitrović and Suraweera (2016) documented long-term deployment of constraint-based tutors for database design (EER-Tutor and KERMIT), demonstrating that CBM can handle large, ill-structured problem spaces where multiple solution paths exist. The constraint representation reduces authoring burden compared to exhaustive procedural models, as constraints capture solution requirements without specifying exact solution sequences. Classroom evaluations showed significant learning gains and high student satisfaction across multiple semesters of deployment.

The primary limitation of CBM lies in constraint engineering: identifying comprehensive, non-redundant constraint sets requires deep domain expertise and iterative refinement. Additionally, CBM provides less granular stepwise guidance than model-tracing approaches, as constraints typically evaluate complete or partial solutions rather than individual problem-solving actions. This makes CBM particularly suitable for design and construction tasks where solution paths

are diverse, but potentially less effective for procedural skill acquisition where step-by-step guidance is beneficial.

3.3 Model-Tracing Cognitive Tutors

Model-tracing cognitive tutors, exemplified by the ACT-R-based tutors developed at Carnegie Mellon University, trace student problem-solving steps against production rule models of expert performance. These systems provide immediate feedback on each action, flag errors as they occur, and offer context-specific hints when students request assistance. The model-tracing approach enables fine-grained diagnosis of student strategies and supports mastery learning through individualized practice sequences.

Tacoma et al. (2020) conducted a controlled evaluation comparing intelligent feedback combining model-tracing and constraint-based elements to simple verification feedback in hypothesis-testing tasks. The intelligent feedback condition (N=163) showed improved task success and reduced errors compared to verification-only feedback (N=151), demonstrating the value of diagnostic feedback. However, the study found limited transfer to follow-up tasks, suggesting that immediate corrective feedback may support performance without necessarily building transferable conceptual understanding.

Fancsali et al. (2021) reported on iterative redesigns of MATHia, a large-scale deployment of cognitive tutor technology, showing that targeted scaffold additions and engagement nudges reduced failure rates on specific workspaces. The interventions were informed by learning analytics identifying common error patterns and disengagement signals, illustrating how data-driven refinement can enhance classical ITS architectures. However, the quasi-experimental evaluation design limits causal claims about intervention effectiveness.

The primary challenge for model-tracing tutors remains authoring cost and brittleness: production rule models must anticipate diverse student strategies, and unanticipated approaches can lead to incorrect feedback or system failures. Additionally, the focus on procedural correctness may inadvertently emphasize surface-level performance over conceptual understanding, particularly if students learn to satisfy the tutor's expectations without developing robust mental models.

3.4 Hybrid and Dialogue-Based Tutors

Hybrid tutors combine multiple paradigms—constraint-based evaluation, model-tracing guidance, natural language dialogue—to leverage complementary strengths. Dialogue-based tutors, in particular, aim to support conceptual understanding through Socratic questioning, explanation elicitation, and collaborative problem-solving, moving beyond purely procedural instruction.

Albacete et al. (2018) developed a dialogue-based tutor that used student model predictions to dynamically adjust question granularity, maintaining conceptual dialogue within the learner's zone of proximal development. The system employed

Bayesian knowledge tracing to estimate concept mastery and selected question specificity accordingly, demonstrating how probabilistic student modeling can inform adaptive dialogue strategies. Empirical evaluation showed that adaptive granularity improved learning efficiency compared to fixed-granularity dialogue.

Recent work has explored integrating large language models into hybrid tutor architectures to reduce authoring burden while maintaining pedagogical structure. Chowdhury et al. (2024) developed MWPTutor, which preserves pedagogical finite-state structures while using LLMs for content generation and state-specific responses. Human evaluations showed that this hybrid design outperformed free-form LLM tutoring on pedagogical quality metrics, suggesting that structured scaffolding can mitigate LLM limitations in educational contexts. The system demonstrates a promising middle ground between high-authoring classical tutors and low-fidelity generative tutors.

The evolution from classical rule-based systems to hybrid architectures reflects a broader trend toward combining symbolic and statistical approaches, leveraging explicit pedagogical knowledge where available while using data-driven methods to handle variability and reduce authoring costs. However, rigorous comparative evaluations measuring learning outcomes across paradigms remain scarce, limiting evidence-based selection of architectural approaches for specific educational contexts.

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4. Knowledge Representation and Graph-Based Learning Systems

Explicit representation of domain knowledge—including concept hierarchies, prerequisite relationships, and curricular alignments—enables interpretable sequencing, targeted remediation, and transparent recommendation in adaptive learning systems. Knowledge graphs and ontologies provide structured frameworks for encoding these relationships, supporting both human interpretation and computational reasoning.

4.1 Educational Knowledge Graphs

Educational knowledge graphs represent concepts, learning resources, and their relationships as nodes and edges in a graph structure, enabling traversal algorithms for path planning, prerequisite checking, and similarity computation. Unlike opaque embedding spaces, knowledge graphs provide explicit, inspectable representations that support interpretability and curricular alignment.

Qiu et al. (2022) developed a Bloom-aligned ontology for mathematics curricula, demonstrating that explicit prerequisite structures and cognitive level annotations enable interpretable path recommendations validated in classroom deployments. The ontology encoded concepts at multiple granularities—topics, subtopics, specific skills—and linked them through prerequisite, part-of, and

related-to relationships. Traversal algorithms identified optimal learning paths that respected prerequisite constraints while adapting to individual student knowledge states.

Guo et al. (2023) implemented RadarDMath, a knowledge graph-driven recommendation system for mathematics education, reporting classroom validation showing improvements in navigability and targeted remediation. The system constructed a domain knowledge graph from curriculum standards and textbook content, then used graph-based similarity measures to recommend exercises aligned with student knowledge gaps. Teacher feedback indicated that the explicit prerequisite structure improved their ability to understand and trust system recommendations compared to black-box collaborative filtering approaches.

The primary advantage of knowledge graph approaches lies in their interpretability and alignment with expert domain models: educators can inspect, validate, and refine graph structures, ensuring that recommendations respect pedagogical principles and curricular standards. However, constructing high-quality educational knowledge graphs remains labor-intensive, requiring domain expertise, curriculum analysis, and iterative refinement.

4.2 Prerequisite Modeling and Curriculum Sequencing

Prerequisite relationships—where mastery of concept A is necessary for learning concept B—form the backbone of curriculum sequencing in adaptive systems. Accurate prerequisite models enable systems to identify knowledge gaps, sequence learning activities to respect dependencies, and avoid presenting material for which students lack necessary foundations.

Dong et al. (2023) developed a multilayer knowledge graph framework that incorporated cognitive diagnostic outputs to produce more relevant exercise recommendations. The framework represented concepts, exercises, and students as heterogeneous nodes, with edges encoding prerequisite relationships, exercise-concept alignments, and student-concept mastery estimates. Graph neural network methods propagated information across layers to generate recommendations that balanced prerequisite satisfaction, difficulty appropriateness, and diversity. Controlled trials showed measurable posttest improvements compared to random and difficulty-matched baselines.

The challenge in prerequisite modeling lies in capturing the nuanced, context-dependent nature of prerequisite relationships: some prerequisites are strict (e.g., understanding variables before solving equations), while others are facilitative but not essential (e.g., geometric intuition for algebraic concepts). Additionally, prerequisite structures may vary across instructional approaches and student populations, requiring flexible, data-informed models rather than rigid expert-defined hierarchies.

4.3 Ontology-Driven Adaptive Systems

Ontologies extend knowledge graphs by incorporating formal semantics, inference rules, and standardized vocabularies, enabling automated reasoning and interoperability across systems. Educational ontologies can represent not only domain concepts but also pedagogical strategies, learning objectives, and assessment criteria, supporting comprehensive adaptive system design.

Abu-Rasheed et al. (2025) explored ontology-driven approaches for educational content organization, demonstrating that formal ontologies enable precise curriculum alignment and support automated content tagging and retrieval. The ontology encoded relationships between learning objectives, content types, cognitive levels, and assessment methods, enabling queries such as "find all resources that address learning objective X at Bloom level Y with assessment type Z." This structured representation supported automated curriculum mapping and gap analysis.

However, the complexity of ontology engineering—requiring expertise in both domain content and formal knowledge representation—limits widespread adoption. Additionally, the rigidity of formal ontologies can hinder adaptation to evolving curricula and diverse instructional contexts. Hybrid approaches that combine lightweight knowledge graphs for core prerequisite structures with flexible metadata schemas for content attributes may offer a pragmatic middle ground.

The integration of knowledge graphs with student modeling and pedagogical reasoning remains an active research area. Systems that combine explicit prerequisite structures (knowledge graphs) with data-driven mastery estimation (knowledge tracing) and adaptive sequencing (reinforcement learning or constraint optimization) represent a promising direction for next-generation adaptive platforms, balancing interpretability, personalization, and scalability.

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5. Student Modeling and Knowledge Assessment

Accurate, real-time estimation of student knowledge states is fundamental to adaptive instruction, enabling systems to select appropriate content difficulty, provide targeted feedback, and sequence learning activities to address knowledge gaps. This section examines the evolution of student modeling approaches from classical probabilistic methods to contemporary deep learning architectures, analyzing their theoretical foundations, predictive capabilities, and practical trade-offs.

5.1 Bayesian Knowledge Tracing and Extensions

Bayesian Knowledge Tracing (BKT), introduced by Corbett and Anderson (1995), models skill acquisition as a Hidden Markov process where each skill

transitions from "unlearned" to "learned" state with some probability after each practice opportunity. The model incorporates four parameters per skill: initial knowledge probability, learning rate, slip probability (knowing but answering incorrectly), and guess probability (not knowing but answering correctly). BKT's interpretable parameters and online update capability made it the dominant student modeling approach in ITS for over two decades.

Corbett and Anderson (1995) demonstrated that BKT-driven mastery learning policies—where students practice each skill until estimated mastery probability exceeds a threshold—produced efficient learning trajectories in cognitive tutor deployments. The model's simplicity enabled real-time inference and transparent decision-making, supporting both system adaptation and teacher interpretation of student progress.

However, BKT's assumptions—skill independence, binary mastery states, time-invariant parameters—limit its expressiveness. Extensions have addressed these limitations through various mechanisms. Yudelson (2016) developed individualized BKT that estimates student-specific parameters, capturing individual differences in learning rates and error patterns. Empirical evaluations showed modest predictive improvements over standard BKT, particularly for students with atypical learning trajectories.

Agarwal et al. (2020) introduced recency-weighted BKT variants that incorporate temporal inflection detection to better model cognitive change over time. The approach identifies periods of rapid learning or forgetting and adjusts parameter estimates accordingly, improving prediction accuracy on datasets with non-stationary learning dynamics. Junhu et al. (2018) developed TD-BKT, which incorporates temporal decay and dynamic parameter adjustment, showing improved fit to student interaction logs compared to standard BKT.

Despite these extensions, BKT's fundamental limitation remains its inability to capture complex dependencies between skills, contextual effects, and rich sequential patterns in student behavior. These limitations motivated the development of more expressive, data-driven modeling approaches.

5.2 Deep Knowledge Tracing and Neural Approaches

Deep Knowledge Tracing (DKT), introduced by Piech et al. (2015), applies recurrent neural networks to model student knowledge states as continuous latent vectors that evolve with each interaction. Unlike BKT's discrete skill-specific states, DKT learns distributed representations that can capture skill relationships, contextual effects, and complex temporal patterns from large interaction logs.

The promise of DKT lies in its capacity to learn from data without requiring explicit skill tagging or parameter engineering, potentially reducing authoring burden and improving predictive accuracy. However, early enthusiasm was tempered by reproducibility challenges and questions about what DKT models

actually learn. Gervet et al. (2020) conducted systematic comparisons showing that DKT's advantages over BKT and logistic regression baselines depend critically on dataset characteristics, preprocessing choices, and evaluation protocols. The study found that DKT excelled on datasets with rich sequential structure and multi-skill items but offered limited advantages on simpler datasets where skill independence assumptions held reasonably well.

Xiong et al. (2016) identified sensitivity to data formatting and multi-skill handling as key factors affecting DKT performance, demonstrating that seemingly minor preprocessing decisions—such as how to represent items assessing multiple skills—could substantially alter model predictions. This sensitivity raised concerns about the robustness and interpretability of purely data-driven approaches.

Subsequent work has focused on enhancing DKT's interpretability and robustness through architectural innovations. Dynamic Key-Value Memory Networks (DKVMN), introduced by Zhang et al. (2017) and extended by subsequent work (2022), employ explicit memory slots for concepts, improving handling of multi-skill items and providing more interpretable attention patterns. The memory architecture separates concept representations (keys) from mastery states (values), enabling the model to explicitly track which concepts are being assessed and how mastery evolves.

Transformer-based knowledge tracing models have emerged as the current state-of-the-art for large-scale deployments. Hong (2023) reported efficient transformer-based knowledge tracing for a personalized language education application, demonstrating scalability to long sequences and large user bases. Li et al. (2024) developed LSTM-BERT hybrid architectures for long-sequence data analysis in intelligent tutoring systems, showing that combining recurrent and attention mechanisms improves both accuracy and computational efficiency.

Efforts to embed cognitive structure into neural architectures aim to balance predictive power with interpretability. Pu et al. (2022) developed EAKT, which embeds structured multi-knowledge-component representations into transformer architectures to make latent states more interpretable while retaining predictive gains. Huang et al. (2021) proposed dynamic knowledge diagnosis integrating cognitive features, showing that explicitly incorporating forgetting curves, learning gains, and prior knowledge into neural models improves prediction stability and interpretability.

5.3 Computerized Adaptive Testing and IRT Models

Item Response Theory (IRT) provides a psychometric framework for modeling relationships between student ability, item difficulty, and response probabilities. Computerized Adaptive Testing (CAT) applies IRT to dynamically select test items that maximize information about student ability, enabling efficient, precise ability estimation with fewer items than fixed-form tests.

Ling et al. (2017) implemented CAT in mathematics assessment contexts, demonstrating that adaptive item selection reduced test length while maintaining measurement precision. The study found that immediate feedback during adaptive testing supported process-oriented learning, though effects on motivation were mixed—some students appreciated the challenge of appropriately difficult items, while others found the absence of easy items discouraging.

Wu et al. (2017) integrated computerized dynamic assessment with immediate feedback, showing that adaptive testing combined with scaffolded support during assessment improved posttest performance compared to static assessment. This approach blurs the boundary between assessment and instruction, using the testing process itself as a learning opportunity.

Tsutsumi et al. (2023) developed Deep-IRT, which combines IRT-style student and item parameterization with deep neural architectures through hypernetworks. The approach separates student ability and item difficulty representations within a deep architecture, retaining interpretable ability/difficulty estimates while improving predictive performance through neural sequence modeling. Empirical evaluations showed that Deep-IRT balanced the interpretability of classical IRT with the expressiveness of deep learning, outperforming both standard IRT and pure DKT on several benchmarks.

5.4 Cognitive Diagnostic Models

Cognitive Diagnostic Models (CDMs) extend IRT by providing fine-grained skill mastery profiles rather than unidimensional ability estimates. CDMs require a Q-matrix that specifies which skills each item assesses, enabling diagnosis of specific knowledge gaps and targeted remediation recommendations.

Song et al. (2022) surveyed cognitive diagnostic approaches in knowledge tracing contexts, noting that CDMs offer diagnostic precision valuable for remediation but require engineered Q-matrices and specialist calibration. The survey highlighted that while CDMs provide theoretically grounded diagnostic frameworks, their practical application is limited by the effort required to construct and validate Q-matrices for large item banks.

Recent work has explored integrating CDM principles into neural architectures. Yin et al. (2023) developed Diagnostic Transformer architectures that incorporate cognitive diagnostic constraints into attention mechanisms, aiming to produce more stable, knowledge-level traces rather than pattern detections. The approach uses contrastive training to encourage the model to distinguish between knowledge states and response patterns, improving diagnostic validity.

Lu et al. (2024) proposed layered knowledge tracing architectures with explicit interpretability layers and user interface translations into educational insights to increase teacher and student trust. The three-layer architecture separates data processing, diagnostic inference, and explanation generation, enabling teachers to inspect diagnostic reasoning and adjust system recommendations. Prototype

evaluations showed improvements in perceived trust and actionable insights compared to black-box neural models.

The tension between predictive accuracy and diagnostic validity remains a central challenge in student modeling. While neural models often achieve superior next-item prediction, their diagnostic validity—whether they accurately reflect evolving knowledge states rather than merely detecting response patterns—requires validation beyond predictive metrics. Hybrid approaches that combine interpretable diagnostic frameworks with expressive neural architectures represent a promising direction, but rigorous evaluation of diagnostic validity through controlled experiments measuring learning outcomes remains limited.

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6. Large Language Models in Educational Contexts

The emergence of large language models (LLMs) based on transformer architectures has introduced new possibilities for natural language tutoring, content generation, and explanation in educational systems. However, the application of LLMs to mathematics education faces significant challenges related to reasoning fidelity, hallucination, and pedagogical alignment.

6.1 Transformer Architectures for Educational NLP

Transformer models, particularly BERT and its variants, have been applied to various educational NLP tasks including learner confusion detection, question classification, and content recommendation. Chanaa and El Faddouli (2020) demonstrated that BERT combined with prerequisite-based ontologies could predict learner confusion in MOOC discussion forums, enabling early support interventions. The approach fine-tuned BERT on labeled forum posts and incorporated ontological prerequisite information to improve classification accuracy, showing that combining neural language models with structured domain knowledge enhances performance on educational tasks.

Hong (2023) reported on efficient transformer-based knowledge tracing deployed in a personalized language education application serving large user populations. The system demonstrated that transformer architectures could scale to long interaction sequences and large user bases while maintaining real-time inference capabilities. The deployment highlighted practical considerations including model compression, caching strategies, and incremental update mechanisms necessary for production systems.

Li et al. (2024) developed LSTM-BERT hybrid architectures specifically for long-sequence data analysis in intelligent tutoring systems, showing that combining recurrent and attention mechanisms improves both accuracy and computational efficiency. The hybrid approach uses LSTM layers to compress long interaction histories into compact representations, then applies BERT-style attention for final predictions, balancing expressiveness and computational cost.

6.2 LLM-Augmented Tutoring Systems

Recent work has explored integrating generative LLMs (GPT-family models) into tutoring workflows to reduce content authoring burden and enable flexible natural language interaction. However, empirical evaluations reveal both promise and significant limitations.

Chowdhury et al. (2024) developed MWPTutor, a hybrid system that preserves pedagogical finite-state structures while using LLMs for content generation and state-specific responses. Human evaluations comparing MWPTutor to free-form LLM tutoring found that the structured hybrid approach outperformed pure LLM tutoring on pedagogical quality metrics including hint appropriateness, error diagnosis accuracy, and avoidance of answer leakage. The findings suggest that explicit pedagogical scaffolding can mitigate LLM limitations, preventing pedagogical drift and maintaining instructional coherence.

Ramanathan and Palaniappan (2024) compared ChatGPT, Bard, and Llama 2 on various tutoring tasks, finding task-dependent strengths rather than one model dominating all tutoring functions. The comparative analysis revealed that different models excelled at different aspects of tutoring: some generated more engaging explanations, others provided more accurate step-by-step solutions, and performance varied substantially across mathematical domains. This heterogeneity suggests that model selection should be task-specific rather than assuming universal superiority of any single model.

Yadav (2023) investigated GPT-4's ability to contextualize mathematical problems to student interests without changing solution intent, finding success in most domains but struggles with certain geometry classes. The study demonstrated that LLMs can generate personalized problem variants that maintain mathematical structure while adapting surface features to student interests, potentially increasing engagement. However, the inconsistent performance across domains highlights the need for verification mechanisms to ensure mathematical equivalence.

Lutgereit and Martinus documented Prof Pi, a GPT-4-based tutoring deployment in underserved contexts, highlighting practical benefits alongside constraints of infrastructure and monitoring. The deployment revealed that while LLMs can provide accessible tutoring in resource-limited settings, concerns about answer leakage, inappropriate content, and lack of curriculum alignment require active monitoring and intervention. The work underscores that technical capability alone is insufficient—deployment contexts, teacher training, and ethical safeguards are equally critical.

6.3 Mathematical Reasoning and Verification Challenges

A persistent challenge in LLM-based mathematics tutoring is ensuring correctness of generated solutions, explanations, and feedback. LLMs frequently produce plausible-sounding but mathematically incorrect responses, particularly for multi-

step algebraic manipulations and symbolic reasoning.

Daheim et al. (2024) demonstrated that adding verifiers for stepwise student solutions improves targeted remediation and reduces hallucinations relative to unverified LLM responses. The verification approach uses symbolic math engines or trained discriminator models to check each solution step, flagging errors and constraining LLM feedback generation to address verified errors. Empirical evaluation showed that verifier-augmented tutoring improved error detection accuracy and reduced instances of incorrect feedback, though verifiers themselves remained imperfect and occasionally missed subtle errors.

The fundamental limitation stems from LLMs' training objective—next-token prediction on text corpora—which does not guarantee mathematical consistency or logical validity. While LLMs can learn statistical patterns in mathematical text, they lack the symbolic reasoning capabilities of computer algebra systems and formal proof assistants. This gap motivates hybrid architectures that combine LLM natural language generation with symbolic verification layers.

Current research directions include integrating symbolic math engines (e.g., SymPy, Mathematica) into LLM tutoring pipelines, training specialized mathematical reasoning models on curated datasets of worked solutions, and developing verification models that can detect mathematical errors in generated text. However, achieving reliable mathematical fidelity at the level required for high-stakes educational applications remains an open challenge.

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7. Retrieval-Augmented Generation for Education

Retrieval-Augmented Generation (RAG) addresses LLM hallucination and knowledge currency limitations by grounding generation in retrieved authoritative documents. In educational contexts, RAG enables systems to generate explanations and feedback anchored in textbooks, curriculum standards, and vetted educational resources, potentially improving factual accuracy and pedagogical alignment.

7.1 Vector-Based Retrieval Approaches

Dense retrieval methods encode documents and queries as vectors in a learned embedding space, enabling semantic similarity search. In educational applications, vector retrieval can identify relevant passages from textbooks, worked examples, and instructional materials to support LLM generation.

Levonian et al. (2023) explored vector-based RAG for educational question answering, finding that retrieval improved factual grounding but introduced trade-offs between faithfulness to source texts and human preference for conversational explanations. The study compared pure LLM generation, retrieval-augmented generation with varying numbers of retrieved passages, and human-written responses. Results showed that RAG reduced factual errors and improved

citation accuracy, but human evaluators sometimes preferred the more fluent, less literal responses of pure LLM generation. This finding highlights a fundamental tension: grounding improves accuracy but may reduce perceived naturalness and engagement.

The challenge in vector retrieval for mathematics education lies in capturing mathematical semantics: standard text embeddings may fail to distinguish mathematically distinct but textually similar expressions (e.g., "x + y" vs. "x * y") or to recognize mathematical equivalence across different notational forms. Specialized embedding models trained on mathematical corpora or augmented with symbolic features may be necessary to achieve adequate retrieval precision.

7.2 Knowledge Graph-Enhanced RAG

Combining knowledge graphs with vector retrieval offers a promising approach to improve both retrieval precision and generation grounding. Knowledge graphs provide explicit prerequisite structures and concept relationships that can guide retrieval granularity and contextual prompt construction.

Xing et al. (2024) developed KG-RAG approaches that use knowledge graph indices to guide retrieval granularity and contextual prompts, showing improved groundedness and fidelity in controlled studies. The system first identifies relevant concepts in the knowledge graph based on the student query, then retrieves passages associated with those concepts and their prerequisites. This two-stage retrieval—concept identification followed by passage retrieval—improves precision by ensuring that retrieved content aligns with the student's current knowledge state and learning objectives.

Abu-Rasheed et al. (2025) explored ontology-enhanced retrieval for educational content, demonstrating that formal ontologies enable precise curriculum alignment and support automated content tagging and retrieval. The approach uses ontological relationships to expand queries (e.g., retrieving content about prerequisite concepts when a student struggles with a dependent concept) and to filter retrieved passages based on cognitive level and instructional context.

The primary advantage of KG-enhanced RAG lies in interpretability and curricular alignment: the knowledge graph provides an explicit, inspectable rationale for retrieval decisions, enabling educators to validate and refine retrieval strategies. However, the approach inherits the knowledge graph construction challenge—building comprehensive, accurate domain ontologies remains labor-intensive and requires ongoing maintenance as curricula evolve.

7.3 Grounding vs. Preference Trade-offs

A recurring finding in RAG research is the tension between grounding accuracy and human preference. Levonian et al. (2023) found that while RAG improved factual fidelity, human evaluators sometimes preferred less grounded, more conversational LLM responses. This preference may reflect several factors:

grounded responses can be overly literal or verbose, retrieval may introduce irrelevant context that distracts from the core question, and humans may value fluency and engagement over strict accuracy in certain educational contexts.

Xing et al. (2024) observed similar trade-offs in KG-RAG evaluations, noting that increased grounding sometimes reduced subjective preference despite improving objective accuracy. The study suggested that optimal grounding strategies may be context-dependent: high-stakes assessment and foundational concept explanation may prioritize accuracy over fluency, while exploratory learning and motivational contexts may tolerate less grounding in favor of engagement.

Resolving this tension requires understanding the pedagogical goals of specific interactions. Systems might adaptively adjust grounding intensity based on context: strict grounding for procedural instruction and assessment, looser grounding for exploratory dialogue and motivation. Additionally, improving the naturalness of grounded generation—through better prompt engineering, retrieval summarization, and generation post-processing—may reduce the preference gap.

The integration of RAG with student modeling and knowledge graphs represents a promising direction for next-generation adaptive tutoring systems. By combining explicit prerequisite structures (knowledge graphs), data-driven mastery estimation (knowledge tracing), and grounded generation (RAG), systems can provide personalized, accurate, and pedagogically sound instruction at scale. However, rigorous evaluation measuring learning outcomes rather than only generation quality remains necessary to validate these approaches.

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8. Multi-Modal Learning Environments and Visualization

Effective STEM education often requires multiple representational formats—symbolic notation, graphical visualizations, interactive simulations, and natural language explanations—to support diverse learning preferences and facilitate conceptual understanding. This section examines the role of visualization and multi-modal interaction in adaptive learning systems.

8.1 Interactive Visualizations and Manipulatives

Dynamic visualizations and interactive manipulatives enable learners to explore mathematical and scientific concepts through direct manipulation, supporting constructivist learning principles. Luu (2025) developed animation libraries using MANIM for mathematical derivations and acoustics concepts, reporting that targeted animations aided comprehension of dynamic processes and abstract relationships. The animations provided step-by-step visual representations of algebraic manipulations, geometric transformations, and physical phenomena, enabling learners to observe processes that are difficult to convey through static text or images alone.

The effectiveness of visualizations depends critically on their integration with instructional scaffolding. Passive viewing of animations provides limited benefit; active engagement—predicting outcomes, manipulating parameters, explaining observed phenomena—is necessary for deep learning. Adaptive systems can enhance visualization effectiveness by prompting prediction before revealing outcomes, asking learners to explain observed patterns, and adjusting visualization complexity based on learner expertise.

8.2 Multi-Representational Workspaces

Multi-representational learning environments present concepts simultaneously in multiple formats—equations, graphs, tables, verbal descriptions—and support coordinated manipulation across representations. Liu et al. (2024) found that multi-representational workspaces combined with adaptive prompts shortened completion time and supported performance improvements in mathematics ITS deployments. The system enabled learners to manipulate algebraic expressions while observing corresponding changes in graphical representations, supporting understanding of relationships between symbolic and visual forms.

Fancsali et al. (2021) documented that engagement nudges and process-oriented prompts in multi-representational mathematics tutors improved engagement with available scaffolds and reduced failure rates. The prompts encouraged learners to use multiple representations strategically—for example, sketching graphs to support algebraic problem-solving or constructing tables to identify patterns—rather than relying exclusively on procedural symbol manipulation.

The design challenge lies in managing cognitive load: multiple simultaneous representations can overwhelm working memory if not carefully coordinated. Effective designs use adaptive scaffolding to introduce representations gradually, provide explicit guidance on when and how to use each representation, and support learners in constructing connections between representational forms.

8.3 Computer Vision and Handwriting Recognition

Integrating computer vision capabilities enables tutoring systems to accept handwritten input, photograph-based problem submission, and gesture-based interaction, bridging the gap between traditional paper-based work and digital tutoring systems. Guerino et al. (2025) developed MathAIde, a teacher-centered system with photo-based exercise correction, emphasizing mixed-methods evaluation to improve classroom adoption in resource-limited settings. The system allows students to photograph handwritten solutions, uses computer vision to extract mathematical expressions, and provides automated feedback on correctness and common errors.

The approach addresses a practical barrier to ITS adoption: many students, particularly in resource-limited contexts, prefer or require paper-based work. By accepting photographed solutions, systems can provide automated feedback without requiring students to learn specialized input interfaces or abandon

familiar work practices. However, handwriting recognition for mathematical notation remains challenging due to variability in notation, ambiguous symbols, and two-dimensional layout. Current systems achieve acceptable accuracy for printed or carefully handwritten expressions but struggle with cursive writing, non-standard notation, and complex layouts.

The integration of multi-modal input and output—text, speech, handwriting, visualization—represents an important direction for inclusive, accessible adaptive learning systems. However, each modality introduces technical challenges and potential failure modes that must be carefully managed through robust error handling, user feedback mechanisms, and graceful degradation when recognition fails.

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9. Integrated Adaptive Learning Platforms

Successful adaptive learning platforms require integration of multiple components—student modeling, content sequencing, feedback generation, visualization, assessment—into coherent, scalable systems. This section examines full-stack architectures, real-time adaptation mechanisms, and teacher-in-the-loop design considerations.

9.1 Full-Stack System Architectures

End-to-end adaptive learning platforms must address data pipelines, model serving, user interface design, content management, and analytics infrastructure. Bystrenina and Nikitin (2022) developed a Markov-based adaptive testing module for Moodle, demonstrating modular implementation and analytics for individual trajectories. The system integrated with existing learning management system infrastructure, enabling deployment without requiring wholesale platform replacement. The modular architecture separated assessment logic from content delivery, enabling independent development and testing of adaptive algorithms.

Dong et al. (2023) implemented multilayer knowledge graph frameworks that incorporated cognitive diagnostic outputs to produce relevant exercise recommendations, reporting measurable posttest improvements in controlled trials. The system architecture comprised offline components (knowledge graph construction, exercise tagging, model training) and online components (real-time student modeling, recommendation generation, feedback delivery), balancing computational efficiency with adaptation responsiveness.

The engineering challenge lies in managing latency, scalability, and reliability: real-time student modeling and recommendation generation must complete within acceptable response times (typically <1 second for interactive feedback), systems must scale to thousands of concurrent users, and failures must degrade gracefully without disrupting learning sessions. Practical deployments employ

caching strategies, model compression, asynchronous processing, and fallback mechanisms to meet these requirements.

9.2 Real-Time Adaptation and Feedback Mechanisms

Effective adaptation requires not only accurate student models but also well-designed decision rules that translate model estimates into instructional actions. Albacete et al. (2018) demonstrated that tying prompt granularity to student model predictions—using Bayesian knowledge tracing estimates to select question specificity—kept conceptual dialogue within learners' zones of proximal development and improved learning efficiency. The decision rules incorporated thresholds and hysteresis to avoid excessive adaptation that could confuse learners with rapidly changing support levels.

Ascari et al. (2021) developed affective and error-based interventions that detected error types and affective states to improve engagement and outcomes in fraction-game experiments. The system monitored both performance indicators (error patterns, response times) and affective signals (inferred from interaction patterns such as rapid clicking, long pauses, help-seeking) to trigger appropriate interventions—cognitive scaffolding for knowledge gaps, motivational messages for disengagement, breaks for frustration.

The challenge in real-time adaptation lies in balancing responsiveness and stability: overly reactive systems may adapt to noise rather than signal, while insufficiently responsive systems fail to address emerging difficulties. Effective designs incorporate temporal smoothing, confidence thresholds, and multi-indicator decision rules to achieve appropriate adaptation rates.

9.3 Teacher-in-the-Loop Design Considerations

Successful classroom deployment requires not only effective student-facing functionality but also teacher-facing tools for monitoring, intervention, and system customization. Guerino et al. (2025) emphasized teacher-centered design in MathAIde, incorporating teacher dashboards, manual override capabilities, and mixed-methods evaluation to improve classroom adoption. The system provided teachers with visualizations of class-wide progress, alerts for struggling students, and tools to review and modify automated feedback before delivery.

Lu et al. (2024) developed layered knowledge tracing architectures with explicit interpretability layers and user interface translations into educational insights to increase teacher trust. The three-layer architecture separated data processing, diagnostic inference, and explanation generation, enabling teachers to inspect diagnostic reasoning and adjust system recommendations. Prototype evaluations showed improvements in perceived trust and actionable insights compared to black-box neural models.

The teacher-in-the-loop approach recognizes that full automation is neither feasible nor desirable in educational contexts: teachers possess contextual knowledge

about individual students, classroom dynamics, and curricular goals that systems cannot capture. Effective designs position technology as a teacher assistant rather than replacement, augmenting teacher capabilities while preserving professional judgment and control.

However, teacher involvement introduces operational complexity: systems must support teacher workflows, provide timely and interpretable information, and avoid creating excessive monitoring burdens. Balancing automation and teacher control remains an active design challenge, requiring iterative co-design with educators and careful attention to usability and integration with existing classroom practices.

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10. Research Gaps and Future Directions

Despite substantial progress in adaptive learning technologies, significant methodological, technical, and practical gaps remain. This section synthesizes identified limitations and articulates priority research directions.

10.1 Methodological and Evaluation Gaps

Limited large-scale randomized controlled trials. The majority of ITS evaluations employ pilot studies, quasi-experimental designs, or within-subjects comparisons, limiting causal claims about effectiveness. Chowdhury et al. (2024), Dong et al. (2023), and Xing et al. (2024) all note the need for rigorous, large-scale randomized evaluations comparing hybrid KG+LLM tutors to classical ITS and standard instruction. Such trials are necessary to establish not only predictive accuracy but actual learning gains, retention, and transfer.

Prediction metrics vs. learning outcomes. Student modeling research predominantly evaluates next-item prediction accuracy (AUC, RMSE) rather than measuring learning outcomes, retention, or transfer. Gervet et al. (2020) and Yin et al. (2023) highlight that high predictive accuracy does not guarantee diagnostic validity or instructional effectiveness. Future work must prioritize end-to-end evaluations that measure learning gains, long-term retention, and transfer to novel problems, not merely model performance on held-out interaction logs.

Lack of standardized benchmarks. Inconsistent preprocessing, evaluation protocols, and dataset characteristics make cross-study comparisons difficult. Xiong et al. (2016) demonstrated that seemingly minor preprocessing decisions substantially affect model performance, yet many studies do not report sufficient methodological detail for replication. The field requires standardized benchmarks with agreed-upon preprocessing, multi-skill handling conventions, and comprehensive evaluation metrics covering both prediction and diagnostic validity.

Insufficient attention to equity and accessibility. Most research focuses on technical performance in well-resourced contexts, with limited examination of deployment challenges in under-resourced settings, accessibility for learners with disabilities, or cultural and linguistic adaptation. Lutgereit and Martinus's Prof Pi deployment and Guerino et al.'s (2025) MathAIde work highlight infrastructure constraints, teacher training needs, and ethical concerns that disproportionately affect underserved populations. Research addressing low-bandwidth deployment, multilingual support, and culturally responsive adaptation is critically needed.

10.2 Technical and Architectural Challenges

Mathematical reasoning fidelity. LLMs exhibit persistent limitations in reliable stepwise mathematical reasoning, symbolic manipulation, and formal verification. Daheim et al. (2024) showed that verifier modules improve targeting but remain imperfect. The integration of symbolic math engines, formal proof assistants, or specialized mathematical reasoning models into tutoring pipelines represents a critical technical challenge. Hybrid architectures that combine LLM natural language capabilities with symbolic verification layers are necessary but not yet mature.

Knowledge graph construction at scale. Building comprehensive, accurate educational knowledge graphs remains labor-intensive, requiring domain expertise, curriculum analysis, and iterative refinement. Abu-Rasheed et al. (2025) and Xing et al. (2024) note that semi-automated methods—LLM-assisted knowledge graph completion, automated prerequisite extraction from textbooks—show promise but require robust human validation workflows and quality metrics. Research developing efficient, validated knowledge graph construction pipelines with human-in-the-loop refinement is essential for scaling graph-based approaches.

Grounding vs. preference optimization. Levonian et al. (2023) and Xing et al. (2024) document the empirical trade-off between textbook-grounded responses and student preferences for conversational explanations. Resolving this tension requires understanding pedagogical goals of specific interactions and developing generation strategies that maintain accuracy while improving naturalness. Research on context-adaptive grounding—adjusting grounding intensity based on instructional goals, student expertise, and interaction type—represents an important direction.

Interpretability and trust. Deep student models achieve high predictive accuracy but lack transparency, limiting teacher trust and adoption. Lu et al. (2024) and Yin et al. (2023) propose architectures with explicit interpretability layers, but comprehensive solutions balancing expressiveness and interpretability remain elusive. Research developing interpretable-by-design architectures, post-hoc explanation methods validated with educators, and user interfaces that effectively communicate model reasoning is critical for practical deployment.

10.3 Equity, Deployment, and Scalability

Infrastructure and connectivity constraints. Many adaptive systems assume reliable high-bandwidth internet connectivity and modern devices, excluding learners in resource-limited contexts. Research on offline-capable systems, low-bandwidth protocols, and progressive enhancement strategies is necessary for equitable access.

Teacher training and professional development. Effective classroom integration requires teacher understanding of system capabilities, limitations, and appropriate use cases. Research on professional development models, teacher-facing documentation, and co-design processes that build teacher capacity and ownership is essential.

Privacy and ethical safeguards. Adaptive systems collect detailed interaction data, raising privacy concerns and potential for misuse. While Lu et al. (2024) address trustworthiness and explainability, the corpus lacks detailed empirical treatments of privacy-preserving techniques, consent protocols, and ethical frameworks for educational data use. Research developing and validating privacy-preserving student modeling, transparent data governance, and ethical guidelines for AI in education is urgently needed.

Longitudinal effectiveness and sustainability. Most evaluations measure immediate learning gains; few examine long-term retention, transfer, or sustained engagement over months or years. Additionally, system maintenance, content updates, and adaptation to evolving curricula require ongoing resources. Research on sustainable deployment models, including cost-effectiveness analyses and long-term impact studies, is necessary to inform institutional adoption decisions.

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11. Conclusion

This comprehensive literature review has synthesized findings from 1,328 peer-reviewed publications spanning foundational learning theories, classical and modern intelligent tutoring systems, knowledge representation frameworks, student modeling approaches, large language model applications, and integrated adaptive learning platforms. The review reveals a field in transition: classical ITS architectures grounded in cognitive theory and explicit domain models are being augmented—and in some cases replaced—by data-driven, neural approaches that offer greater scalability and flexibility but introduce new challenges related to interpretability, mathematical fidelity, and pedagogical alignment.

Several key themes emerge from this synthesis. First, theoretical foundations—cognitive load theory, zone of proximal development, information processing, and self-regulated learning frameworks—provide essential design principles that remain relevant despite technological evolution. Successful adaptive systems navigate inherent tensions between these frameworks through context-sensitive

design, real-time student modeling, and empirical validation. Second, the evolution from rule-based tutors to hybrid architectures reflects a broader trend toward combining symbolic and statistical approaches, leveraging explicit pedagogical knowledge where available while using data-driven methods to handle variability and reduce authoring costs. Third, knowledge graphs and ontologies enable interpretable sequencing and transparent recommendation, addressing limitations of opaque embedding spaces and supporting curricular alignment. Fourth, student modeling has progressed from simple probabilistic trackers to sophisticated neural architectures, but the field must prioritize diagnostic validity and learning outcomes over predictive accuracy alone. Fifth, large language models introduce powerful natural language capabilities but require verification layers, grounding mechanisms, and pedagogical scaffolding to ensure mathematical correctness and instructional coherence. Sixth, retrieval-augmented generation offers a promising approach to ground AI-generated content in authoritative resources, though trade-offs between grounding accuracy and human preference require careful navigation.

Critical research gaps persist across methodological, technical, and practical dimensions. Methodologically, the field requires large-scale randomized controlled trials measuring learning outcomes rather than only predictive accuracy, standardized benchmarks enabling fair cross-study comparisons, and greater attention to equity, accessibility, and deployment in under-resourced contexts. Technically, achieving reliable mathematical reasoning in generative models, scaling knowledge graph construction, balancing interpretability and expressiveness in student models, and resolving grounding-preference trade-offs remain open challenges. Practically, addressing infrastructure constraints, supporting teacher professional development, ensuring privacy and ethical data use, and demonstrating long-term effectiveness and sustainability are essential for widespread adoption.

The proposed research on intelligent adaptive learning platforms for numerical analysis and scientific computing is well-positioned to address these gaps through hybrid architectures that combine knowledge graph-guided retrieval, verifier-augmented LLM generation, and interpretable student modeling. By integrating explicit prerequisite structures with data-driven personalization, symbolic verification with natural language generation, and teacher-in-the-loop design with automated adaptation, such platforms can advance both theoretical understanding and practical deployment of AI-enhanced STEM education. However, success requires not only technical innovation but also rigorous evaluation, educator collaboration, and sustained attention to equity and accessibility. The convergence of educational theory, knowledge representation, deep learning, and generative AI creates unprecedented opportunities to realize the long-standing vision of personalized, effective, scalable STEM education—but only if the field addresses identified gaps with methodological rigor, technical sophistication, and ethical responsibility.

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